

## Supplementary Information (SI) Appendix

“Yeast metabolic and signaling genes are required for heat shock survival and have little overlap with the heat-induced genes”

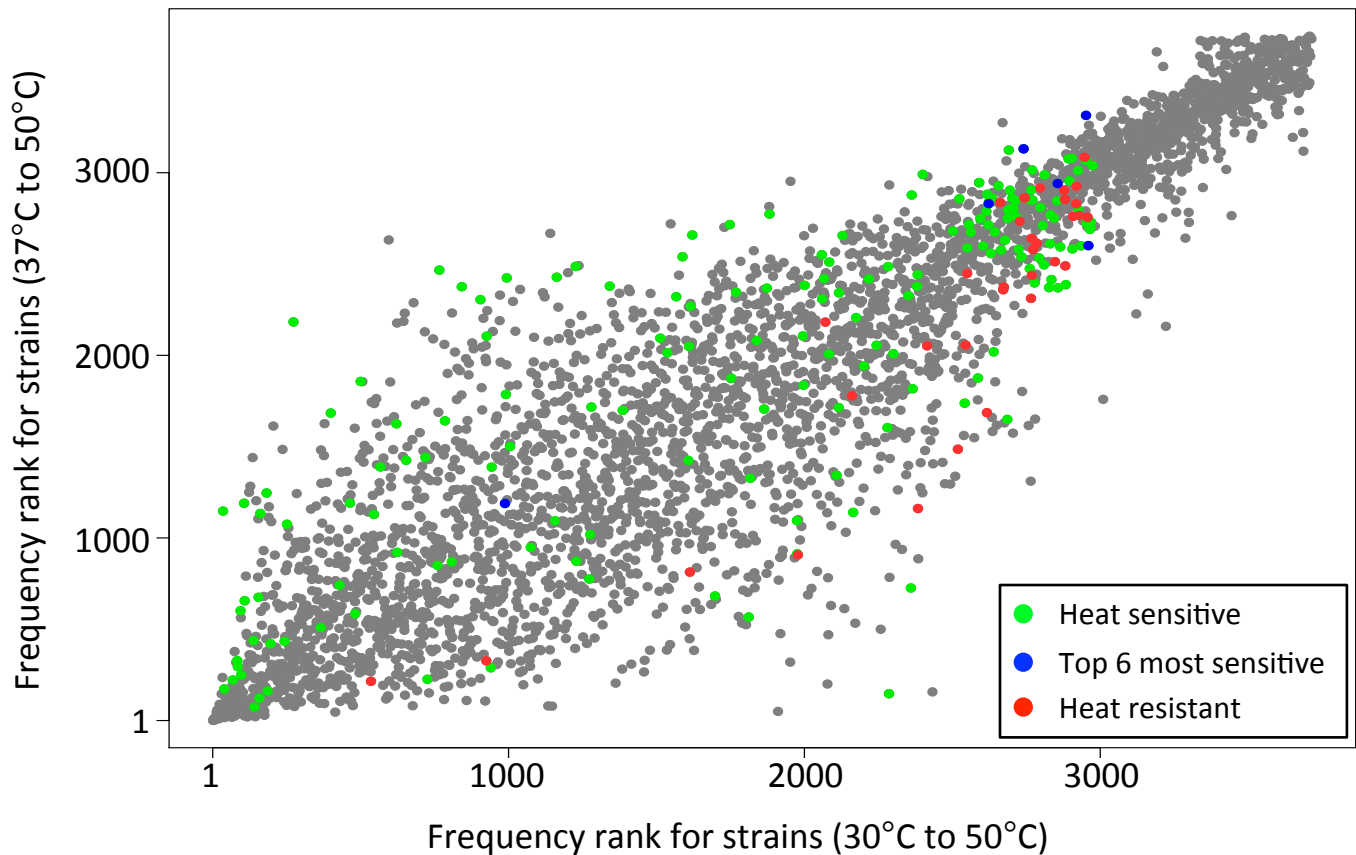
Gibney, P.A., Lu, C., Caudy, A.A., Hess, D.C., and Botstein, D.

### Supplementary Figures:

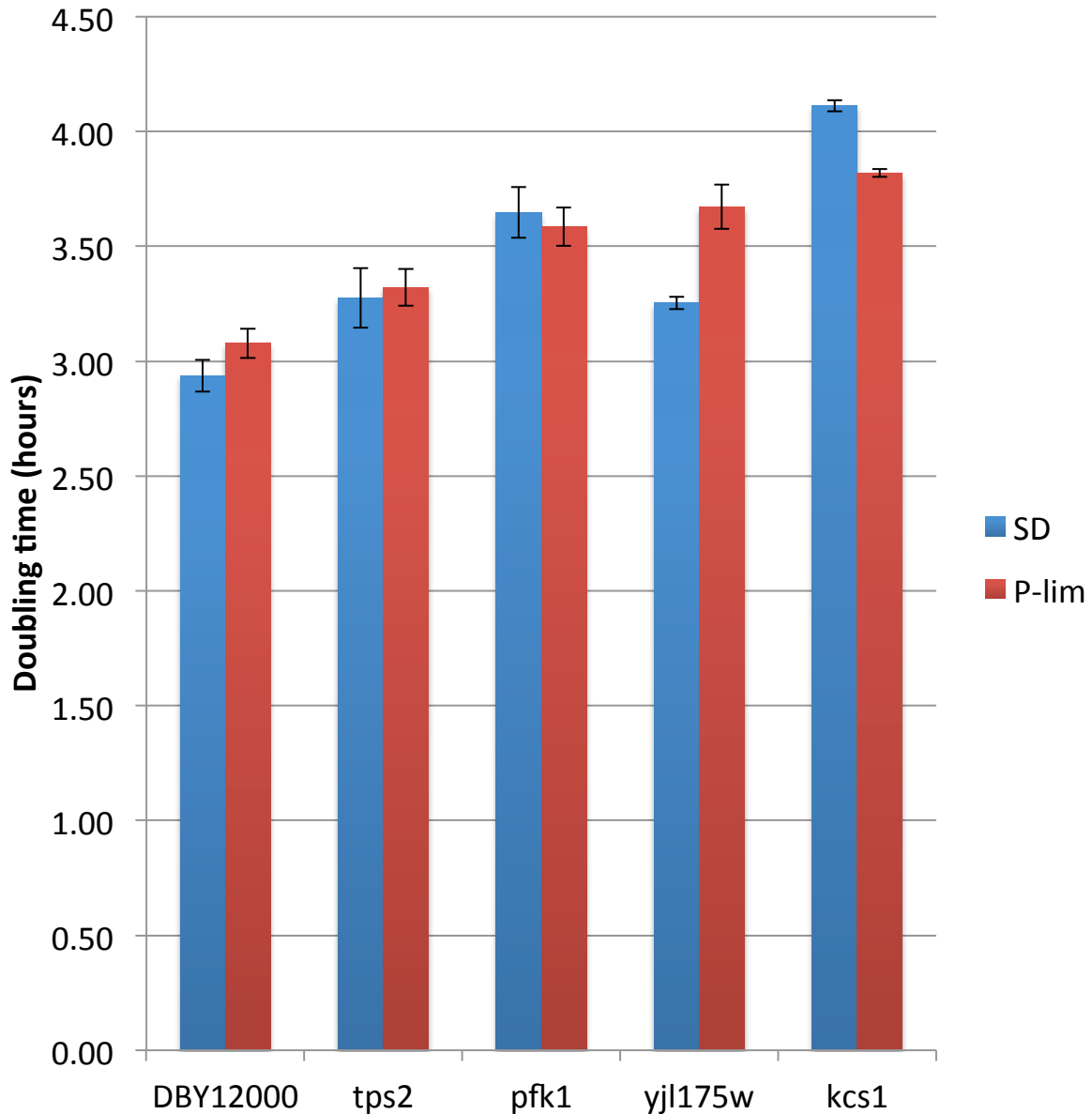
- Figure S1. Initial populations are generally similar at time zero between two different experiments, and extreme death rates are not related to initial strain abundance.
- Figure S2. Doubling times of most sensitive deletion strains are not significantly different from wild type.
- Figure S3. Independent validation of screen top hits.
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- Figure S5. Comparison of chaperone protein heat shock induction versus strain abundance.
- Figure S6. Examination of gene expression after lethal heat pulse.
- Figure S7. Plating for survival on non-fermentable carbon source does not repair *pfk1Δ* thermosensitivity.
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### Supplementary Tables:

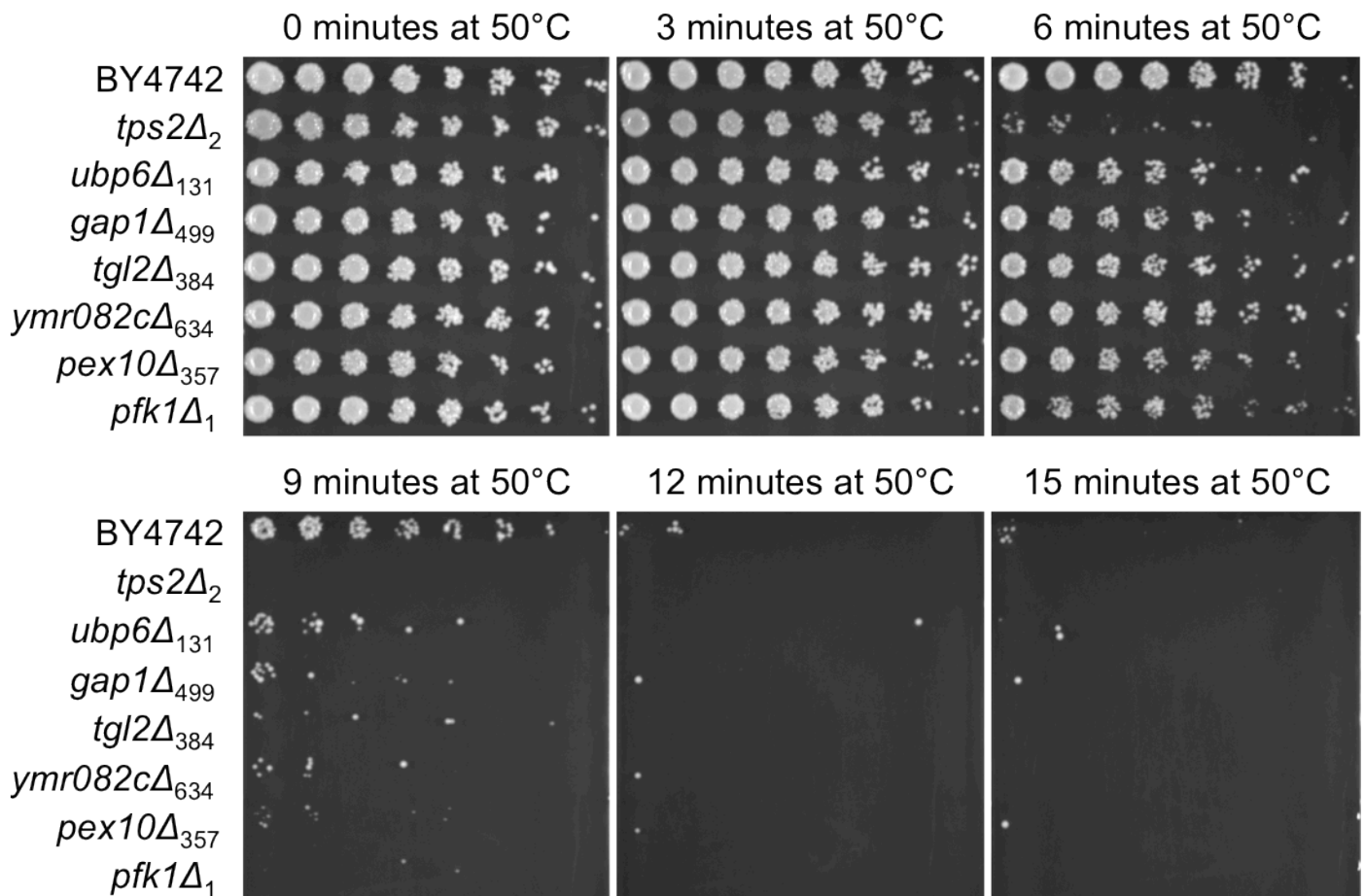
- Table S1. Strains filtered-out of 30°C - 50°C results with significant death rates ( $p < 0.01$ ) due to low CFUs at time zero.
- Table S2. Major chaperone proteins in *S. cerevisiae*.
- Table S3. Strains filtered-out of 37°C - 50°C results with significant death rates ( $p < 0.01$ ) due to low CFUs at time zero.
- Table S4. 205 gene deletion strains present in MATa yeast deletion collection from OpenBiosystems (version 5), but absent in constructed prototrophic collection (due to auxotrophy, sterility, sporulation defects, or reasons unknown)
- Table S5. Chemostat media components.
- Table S6. Colonies collected at each time point.
- Table S7. Barcode mapping and data filtering statistics.
- Table S8. GO Term finder analysis of gene sets



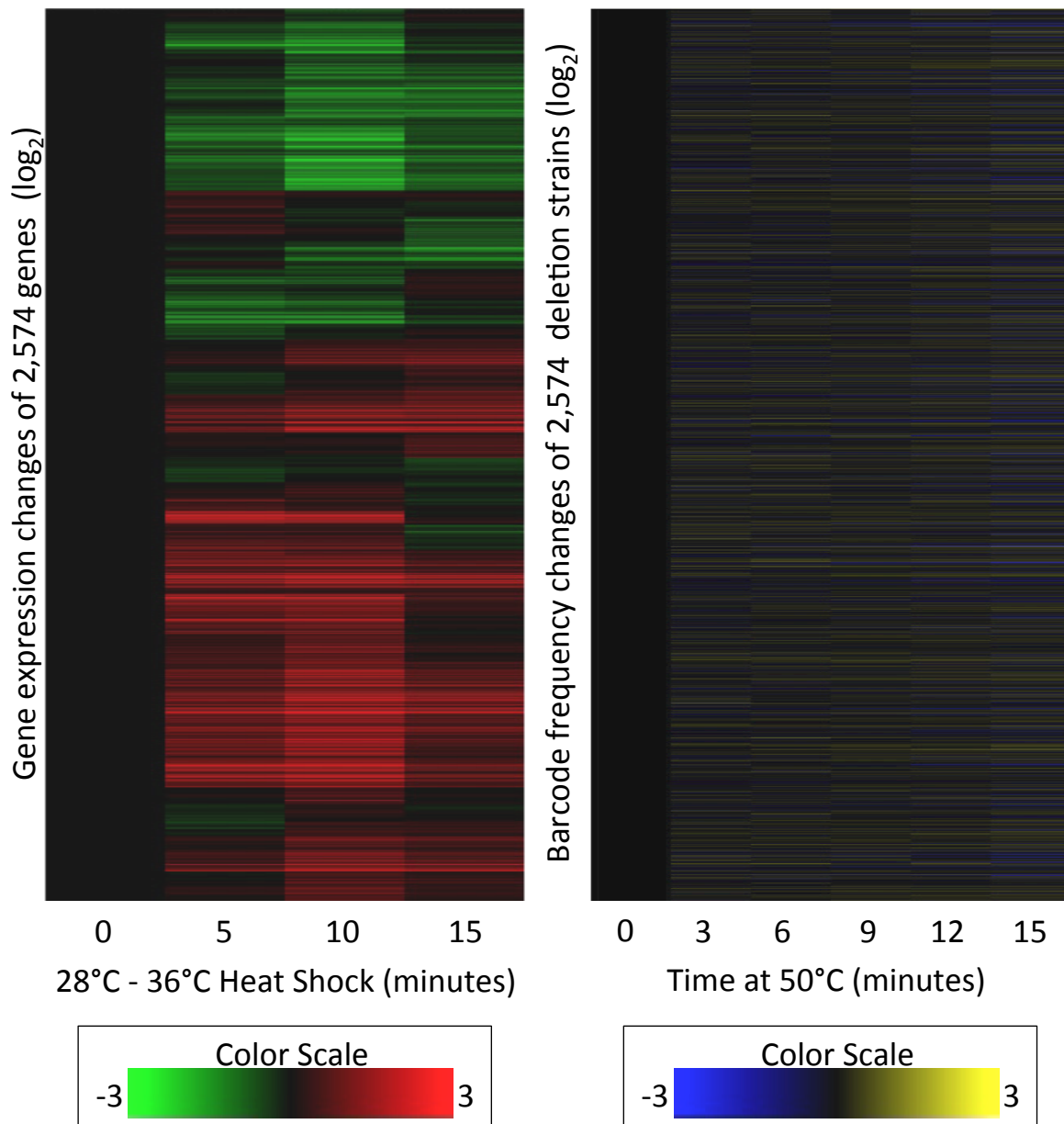
**Figure S1. Initial populations are generally similar at time zero between two different experiments, and extreme death rates are not related to initial strain abundance.** Genes were ranked in order of barcode frequency within the population for both 30°C-50°C and 37°C-50°C experiment (rank of 1 is most abundant). Frequency ranks were then plotted against each other using a scatter plot to examine reproducibility of gene frequency within the starting population ( $t = 0$  min) between experiments. Only genes with data for both data sets are shown. Fitted line equation:  $y = 0.9811x$ ,  $r^2 = 0.8431$ . Gene deletions that exhibit most extreme ( $p < 0.05$ ) heat sensitivity and heat resistance are colored as indicated in legend.



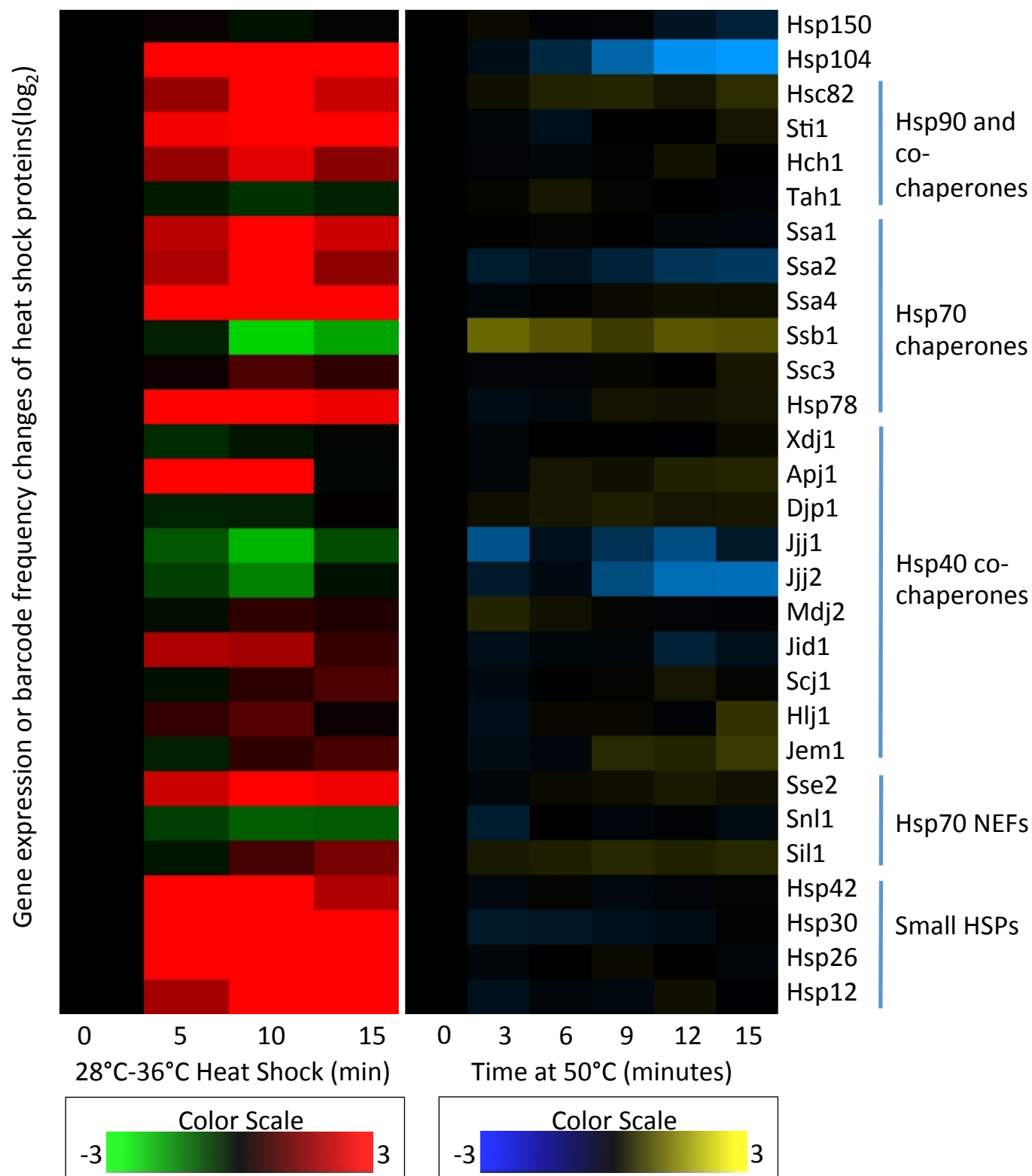
**Figure S2. Doubling times of most sensitive deletion strains are not significantly different from wild type cells.** Doubling times were calculated for wild type (DBY12000) and individual deletion strains grown in biological triplicate in the indicated media in a BioTek Synergy H1 (P-lim indicates phosphate-limited chemostat media, SD indicates YNB + 2% glucose) at 30°C. Error bars represent the standard deviation of three biological replicates.



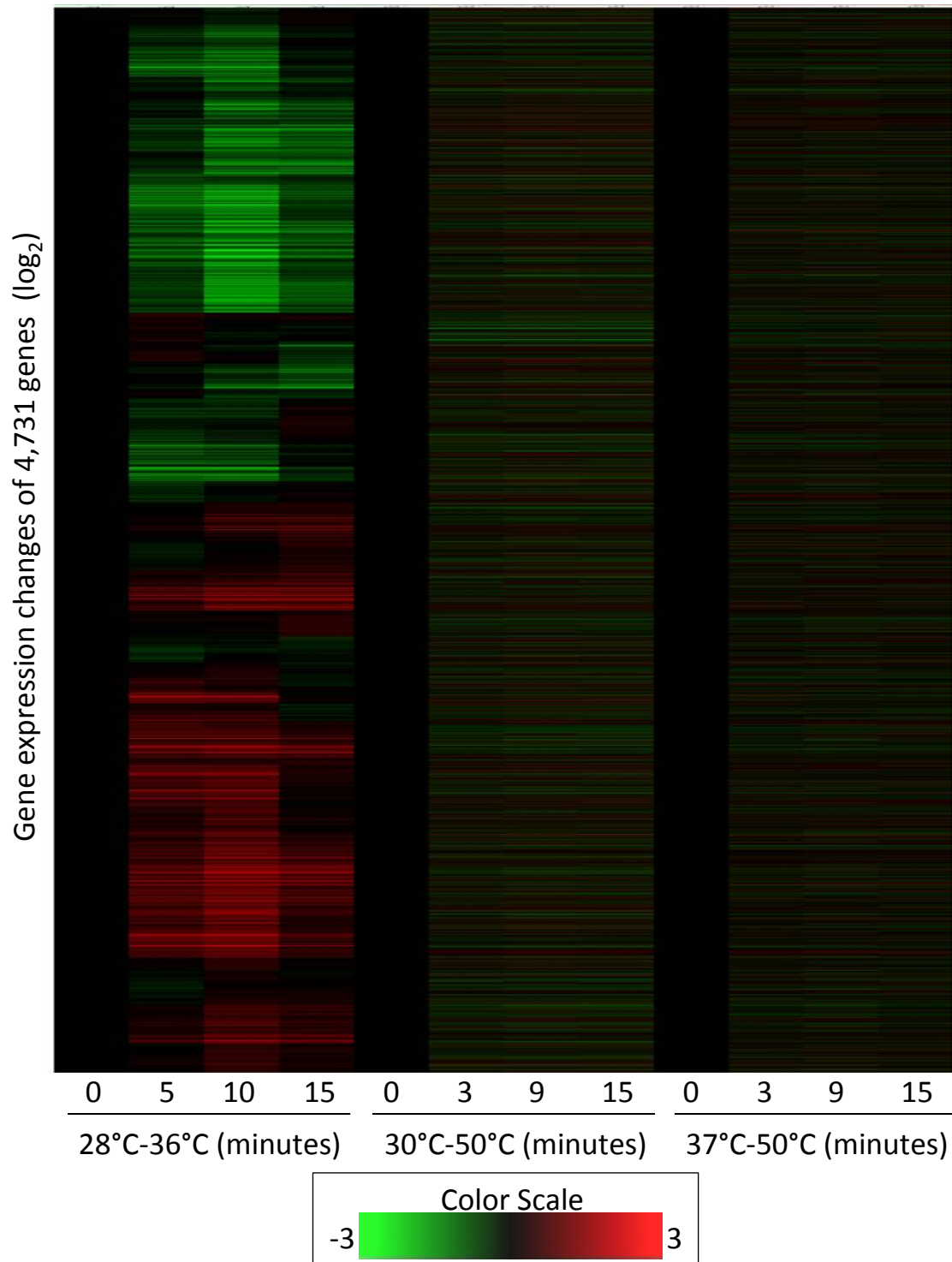
**Figure S3. Independent confirmation of screen top hits.** Gene deletions were re-made in a wild type heterozygous diploid, confirmed, then sporulated to obtain haploid deletion strains. Strains were grown to log phase in SD media, then heat shocked at 50°C for the indicated amount of time. At the indicated time, the sample was moved to ice. After the time-course, all samples were serially diluted into fresh YPD (1/3 dilution steps) and spotted onto YPD plates to qualitatively assess survival. Strains selected all had negative death rates, with values ranging from extremely significant to less significant (in subscript next to each gene is the rank in order of most negative death rate to least negative death rate out of the entire data set (3877 strains)).



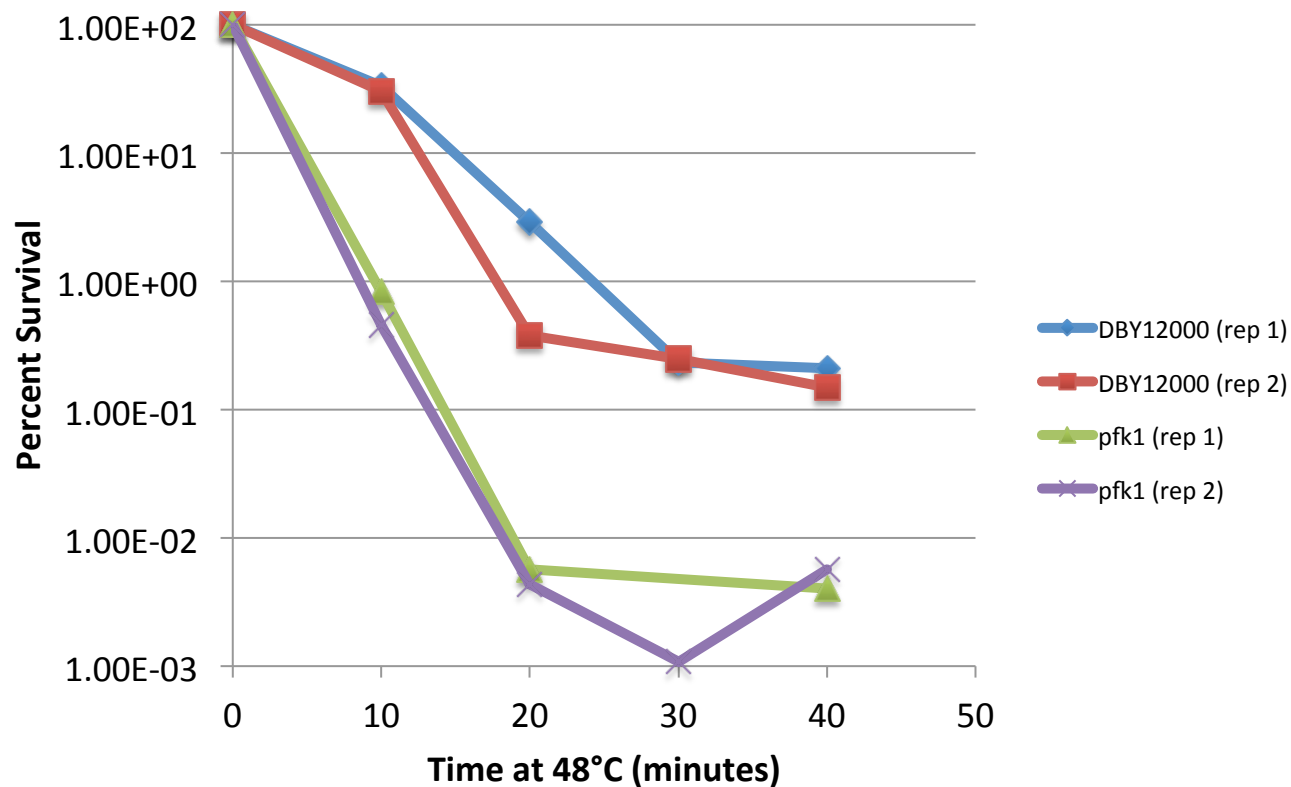
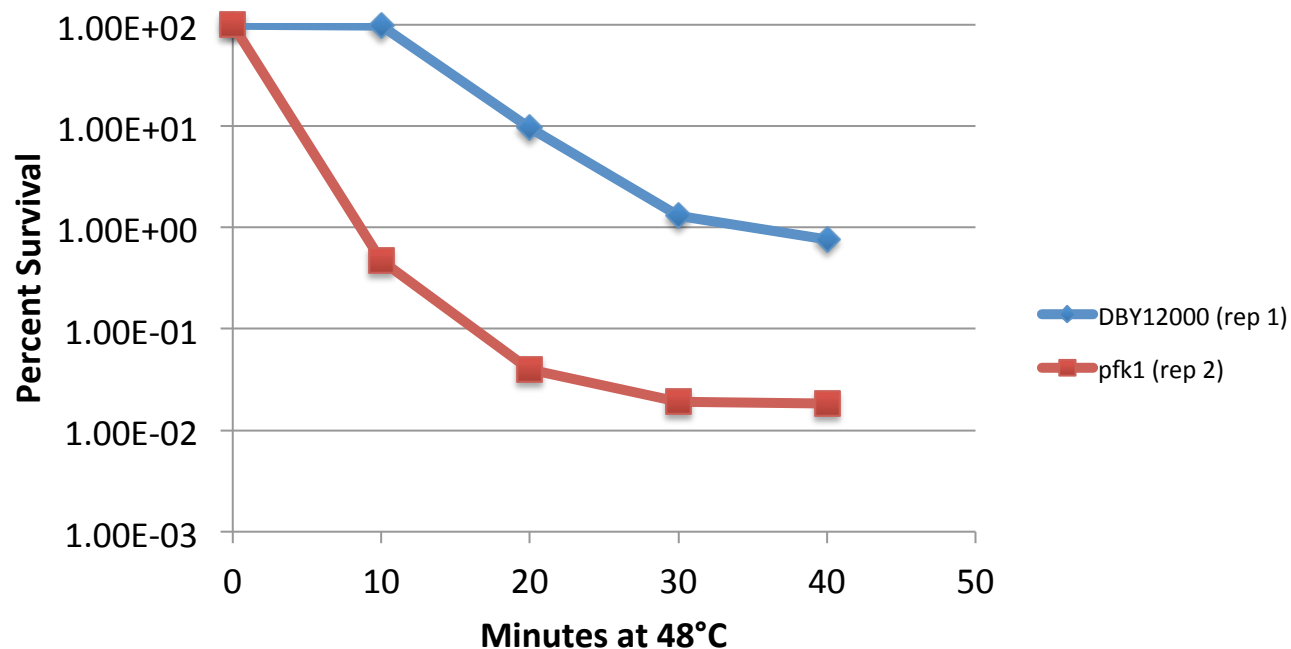
**Figure S4. Deletion strains most sensitive to lethal heat shock (30°C to 50°C) do not correlate with specific gene expression responses to a mild heat shock (shift from 28°C to 36°C).** LEFT: Gene expression changes for 2,574 genes after a mild heat shock for 0, 5, 10, and 15 minutes hierarchically clustered using Pearson correlation. RIGHT: Barcode frequency changes for the cognate gene deletions is shown (gene order is preserved between analyses). Color scale indicates log<sub>2</sub>-transformed fold changes.



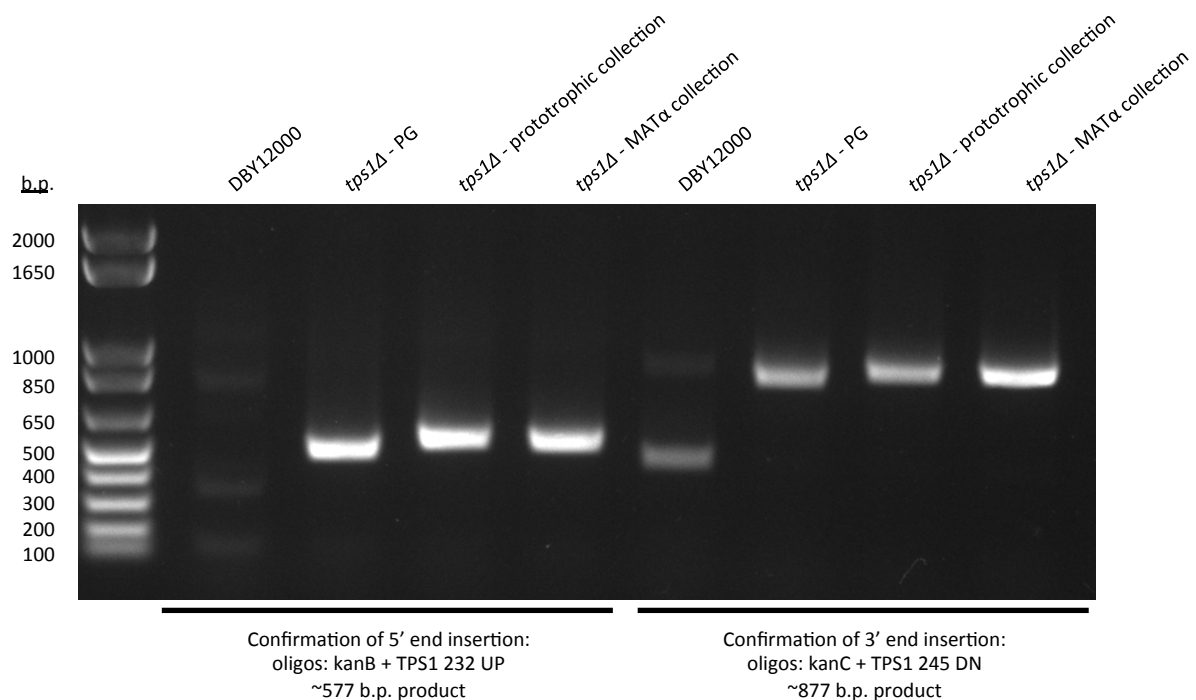
**Figure S5. Comparison of chaperone protein heat shock induction versus strain abundance.** Gene expression (red/green) and barcode frequency changes (yellow/blue) are shown for major classes of heat shock protein chaperones (and co-chaperones). This figure does not contain genes that have no barcode frequency change data (either because the gene is essential, was not detected, or was filtered-out of the dataset due to low numbers at t=0). See SI Appendix, tab. S2 for the source list of chaperone genes.



**Figure S6. Examination of gene expression after lethal heat pulse.** Gene expression changes (red/green) are shown for cultures shifted to the indicated temperatures for the indicated times. Color scale represents  $\log_2$  changes in expression.

**A.****B.**

**Figure S7. Plating for survival on non-fermentable carbon source does not repair *pfk1*Δ thermosensitivity.** Wild type (DBY12000) and *pfk1*Δ cells were grown to log phase in SD media, then heat shocked at 48°C and sampled at the indicated time points. Samples were plated for viability onto either (A) YPD or (B) YPGE media.



### Figure S8. PCR confirmation of *tps1Δ* deletion strains.

PCR confirmation of all three *TPS1* deletions was performed at both the 5' and 3' ends using primers indicated.

TPS1 232 UP: 5' - gcgccatggacaaactgcactgag - 3'

TPS1 245 DN: 5' - cgaagaagagatcagcgcgggag - 3'

**Table S1: Strains filtered-out of 30°C - 50°C results with significant death rates ( $p < 0.01$ ) due to low CFUs at time zero**

| ORF     | Name    | Death Rate | CFU (t = 0 minutes) |
|---------|---------|------------|---------------------|
| YKL081W | TEF4    | -23.11     | 10.22               |
| YOL081W | IRA2    | -18.49     | 17.51               |
| YIL128W | MET18   | -16.62     | 16.59               |
| YJL189W | RPL39   | -15.96     | 23.04               |
| YDR458C | HEH2    | -14.79     | 1.52                |
| YCR006C | YCR006C | -14.76     | 5.22                |
| YLR296W | YLR296W | -14.66     | 18.74               |
| YMR141C | YMR141C | -14.27     | 9.21                |
| YDR392W | SPT3    | -14.18     | 2.15                |
| YMR030W | RSF1    | -14.15     | 5.87                |
| YGR153W | YGR153W | -13.5      | 35.01               |
| YIR002C | MPH1    | -13.32     | 25.19               |
| YGL006W | PMC1    | -13.12     | 1.74                |
| YOL128C | YGK3    | -12.95     | 3.48                |
| YBR044C | TCM62   | -12.8      | 0.65                |
| YBR224W | YBR224W | -12.58     | 28.26               |
| YMR026C | PEX12   | -11.25     | 1.3                 |
| YJR047C | ANB1    | -11.21     | 10.44               |
| YPL035C | YPL035C | -10.82     | 0.92                |
| YNL226W | YNL226W | -10.68     | 27.33               |
| YER145C | FTR1    | -10.59     | 3.99                |
| YDR523C | SPS1    | -10.52     | 3.26                |
| YKL085W | MDH1    | -10.51     | 2.39                |
| YLR018C | POM34   | -10.47     | 1.52                |
| YDR209C | YDR209C | -10.44     | 26.42               |
| YHR111W | UBA4    | -10.33     | 26.72               |
| YDL243C | AAD4    | -10.19     | 1.84                |
| YPL073C | YPL073C | -10.13     | 11.06               |
| YGR197C | SNG1    | -10.05     | 6.14                |
| YOL001W | PHO80   | -9.87      | 8.6                 |
| YCR010C | ADY2    | -9.59      | 1.3                 |
| YHL009C | YAP3    | -9.49      | 7.06                |
| YLR384C | IKI3    | -9.36      | 23.35               |
| YLR281C | YLR281C | -8.89      | 11.36               |
| YNL022C | YNL022C | -8.82      | 4.3                 |
| YOR286W | YOR286W | -8.81      | 1.74                |
| YDR213W | UPC2    | -8.69      | 23.65               |
| YOL009C | MDM12   | -8.57      | 10.14               |

|           |         |       |       |
|-----------|---------|-------|-------|
| YOR344C   | TYE7    | -8.51 | 9.21  |
| YKL096W-A | CWP2    | -8.43 | 8.29  |
| YDR368W   | YPR1    | -8.35 | 27.95 |
| YOR304W   | ISW2    | -8.24 | 11.36 |
| YDL065C   | PEX19   | -8.22 | 4.61  |
| YGL248W   | PDE1    | -8.21 | 2.39  |
| YHR013C   | ARD1    | -8.05 | 1.96  |
| YHL011C   | PRS3    | -8.03 | 3.38  |
| YML018C   | YML018C | -7.97 | 2.61  |
| YGR104C   | SRB5    | -7.94 | 30.1  |
| YDR357C   | YDR357C | -7.86 | 22.12 |
| YEL037C   | RAD23   | -7.76 | 1.23  |
| YBL042C   | FUI1    | -7.67 | 27.95 |
| YMR124W   | YMR124W | -7.61 | 7.37  |
| YBR084C-A | RPL19A  | -7.6  | 29.8  |
| YLR109W   | AHP1    | -7.47 | 32.25 |
| YDR336W   | YDR336W | -7.46 | 3.07  |
| YPL219W   | PCL8    | -7.44 | 0.65  |
| YER095W   | RAD51   | -7.3  | 1.3   |
| YER150W   | SPI1    | -7.17 | 3.04  |
| YDL129W   | YDL129W | -7.14 | 1.23  |
| YLR312W-A | MRPL15  | -7.04 | 13.82 |
| YBR077C   | SLM4    | -6.97 | 28.91 |
| YOR044W   | IRC23   | -6.92 | 1.23  |
| YML024W   | RPS17A  | -6.85 | 9.13  |
| YIR025W   | MND2    | -6.85 | 3.99  |
| YFL036W   | RPO41   | -6.77 | 14.44 |
| YPL265W   | DIP5    | -6.76 | 6.45  |
| YGR081C   | SLX9    | -6.7  | 32.25 |
| YER087W   | AIM10   | -6.55 | 3.99  |
| YDR217C   | RAD9    | -6.43 | 0.92  |
| YOR058C   | ASE1    | -6.4  | 20.87 |
| YEL064C   | AVT2    | -6.33 | 4.78  |
| YBL079W   | NUP170  | -6.33 | 3.99  |
| YOR155C   | ISN1    | -6.29 | 4.13  |
| YDR100W   | TVP15   | -6.28 | 4.3   |
| YKL168C   | KKQ8    | -6.28 | 0.43  |
| YKL115C   | YKL115C | -6.21 | 4.92  |
| YCR067C   | SED4    | -6.19 | 0.92  |
| YLR325C   | RPL38   | -6.18 | 0.61  |
| YIL066C   | RNR3    | 6.3   | 6.14  |
| YJL204C   | RCY1    | 6.36  | 10.14 |

|           |           |       |       |
|-----------|-----------|-------|-------|
| YNL177C   | MRPL22    | 6.39  | 8.29  |
| YML017W   | PSP2      | 6.41  | 1.74  |
| YDR159W   | SAC3      | 6.45  | 10.22 |
| YLR431C   | ATG23     | 6.48  | 7.61  |
| YCR098C   | GIT1      | 6.64  | 0.43  |
| YOR073W   | SGO1      | 6.7   | 0.22  |
| YKR067W   | GPT2      | 6.71  | 0.22  |
| YDR347W   | MRP1      | 6.74  | 0.92  |
| YDR028C   | REG1      | 6.9   | 1.09  |
| YMR058W   | FET3      | 7.24  | 5     |
| YPR118W   | MRI1      | 7.3   | 1.09  |
| YPR022C   | YPR022C   | 7.47  | 0.22  |
| YER091C-A | YER091C-A | 7.53  | 5.84  |
| YOR002W   | ALG6      | 7.62  | 33.04 |
| YDR155C   | CPR1      | 7.66  | 0.22  |
| YDR124W   | YDR124W   | 7.81  | 0.22  |
| YMR250W   | GAD1      | 7.83  | 11.67 |
| YMR222C   | FSH2      | 8     | 14.56 |
| YLR405W   | DUS4      | 8.12  | 0.22  |
| YLR269C   | YLR269C   | 8.14  | 0.22  |
| YHR021C   | RPS27B    | 8.48  | 10.75 |
| YDR385W   | EFT2      | 8.48  | 0.31  |
| YGR272C   | YGR272C   | 8.56  | 9.83  |
| YOR049C   | RSB1      | 8.66  | 1.54  |
| YMR191W   | SPG5      | 8.67  | 12.59 |
| YLR173W   | YLR173W   | 8.87  | 2.15  |
| YOR211C   | MGM1      | 9.17  | 1.84  |
| YPL078C   | ATP4      | 9.67  | 0.43  |
| YOL004W   | SIN3      | 9.75  | 3.38  |
| YJL055W   | YJL055W   | 10.37 | 0.61  |
| YCL010C   | SGF29     | 11.04 | 7.99  |
| YPL106C   | SSE1      | 11.44 | 0.61  |
| YLR146C   | SPE4      | 12.18 | 2.39  |
| YPR074C   | TKL1      | 13.96 | 0.87  |
| YKL204W   | EAP1      | 15.21 | 0.92  |

**Table S2: Major chaperone proteins in *S. cerevisiae***

| CATEGORY                | NAME   | ORF            | FUNCTION                                                                                                                      |
|-------------------------|--------|----------------|-------------------------------------------------------------------------------------------------------------------------------|
|                         | Hsp150 | <i>YJL159W</i> | O-mannosylated heat shock protein that is secreted and covalently attached to the cell wall; required for cell wall stability |
| Hsp100-type             | Hsp104 | <i>YLL026W</i> | Heat shock protein that cooperates with Hsp40 and to refold and reactivate previously denatured, aggregated proteins          |
| Hsp90 and co-chaperones | Hsc82  | <i>YMR186W</i> | Cytoplasmic chaperone of the Hsp90 family                                                                                     |
|                         | Hsp82  | <i>YPL240C</i> | Cytoplasmic chaperone of the Hsp90 family                                                                                     |
|                         | Sti1   | <i>YOR027W</i> | Hsp90 co-chaperone; homolog of mammalian Hop                                                                                  |
|                         | Sba1   | <i>YKL117W</i> | Hsp90 co-chaperone; homolog of mammalian p23                                                                                  |
|                         | Cpr6   | <i>YLR216C</i> | Peptidyl-prolyl cis-trans isomerase (cyclophilin)                                                                             |
|                         | Cpr7   | <i>YJR032W</i> | Peptidyl-prolyl cis-trans isomerase (cyclophilin)                                                                             |
|                         | Cns1   | <i>YBR155W</i> | TPR-containing co-chaperone                                                                                                   |
|                         | Aha1   | <i>YDR214W</i> | Co-chaperone that binds to Hsp82p and activates its ATPase activity                                                           |
|                         | Hch1   | <i>YNL281W</i> | Heat shock protein regulator that binds to Hsp90p and may stimulate ATPase activity                                           |
|                         | Tah1   | <i>YCR060W</i> | HSP90 cofactor; contains a single TPR domain with at least two TPR motifs                                                     |
|                         | Cdc37  | <i>YDR168W</i> | Essential Hsp90p co-chaperone; necessary for passage through the START phase of the cell cycle                                |
| Hsp70 (DnaK)            | Ssa1   | <i>YAL005C</i> | Cytosolic Hsp70                                                                                                               |
|                         | Ssa2   | <i>YLL024C</i> | Cytosolic Hsp70                                                                                                               |
|                         | Ssa3   | <i>YBL075C</i> | Cytosolic Hsp70                                                                                                               |
|                         | Ssa4   | <i>YER103W</i> | Cytosolic Hsp70                                                                                                               |
|                         | Ssb1   | <i>YDL229W</i> | Ribosome-associated Hsp70                                                                                                     |
|                         | Ssb2   | <i>YNL209W</i> | Ribosome-associated Hsp70                                                                                                     |
|                         | Kar2   | <i>YJL034W</i> | Luminal ER Hsp70                                                                                                              |
|                         | Ssc1   | <i>YJR045C</i> | Mitochondrial Hsp70                                                                                                           |
|                         | Ssc2   | <i>YLR369W</i> | Mitochondrial Hsp70                                                                                                           |
|                         | Ssc3   | <i>YEL030W</i> | Mitochondrial Hsp70                                                                                                           |
|                         | Ssz1   | <i>YHR064C</i> | Hsp70 protein that interacts with Zuo1 and Ssb Hsp70s                                                                         |
|                         | Hsp78  | <i>YDR258C</i> | Oligomeric mitochondrial matrix chaperone that cooperates with Ssc1p in mitochondrial thermotolerance after heat shock        |
|                         | Hsp60  | <i>YLR259C</i> | Tetradecameric mitochondrial chaperonin                                                                                       |
| Hsp40 (DnaJ)            | Ydj1   | <i>YNL064C</i> | Regulates Hsp70 ATPase activity; member of DnaJ family                                                                        |
|                         | Xdj1   | <i>YLR090W</i> | Regulates Hsp70 ATPase activity; member of DnaJ family; present in mitochondria                                               |
|                         | Apj1   | <i>YNL077W</i> | Regulates Hsp70 ATPase activity; member of DnaJ family; present in mitochondria                                               |
|                         | Sis1   | <i>YNL007C</i> | Regulates Hsp70 ATPase activity; member of DnaJ family                                                                        |

|            |       |                |                                                                                                               |
|------------|-------|----------------|---------------------------------------------------------------------------------------------------------------|
|            | Djp1  | <i>YIR004W</i> | Regulates Hsp70 ATPase activity; member of DnaJ family; required for peroxisomal protein import               |
|            | Zuo1  | <i>YGR285C</i> | Ribosome-associated DnaJ protein                                                                              |
|            | Swa2  | <i>YDR320C</i> | Auxilin-like protein involved in vesicular transport; member of DnaJ family                                   |
|            | Jjj1  | <i>YNL227C</i> | Regulates Hsp70 ATPase activity; member of DnaJ family; associated with ribosome                              |
|            | Jjj2  | <i>YJL162C</i> | Protein of unknown function, contains a J-domain                                                              |
|            | Jjj3  | <i>YJR097W</i> | Protein of unknown function, contains a J-domain                                                              |
|            | Caj1  | <i>YER048C</i> | Regulates Hsp70 ATPase activity; nuclear member of DnaJ family                                                |
|            | Cwc23 | <i>YGL128C</i> | Component of a complex containing Cef1p, putatively involved in pre-mRNA splicing; contains a J-domain        |
|            | Mdj1  | <i>YFL016C</i> | Regulates Hsp70 ATPase activity; mitochondrial member of DnaJ family                                          |
|            | Mdj2  | <i>YNL328C</i> | Regulates Hsp70 ATPase activity; mitochondrial member of DnaJ family                                          |
|            | Pam18 | <i>YLR008C</i> | Regulates Hsp70 ATPase activity; mitochondrial member of DnaJ family                                          |
|            | Jac1  | <i>YGL018C</i> | Regulates Hsp70 ATPase activity; mitochondrial member of DnaJ family                                          |
|            | Jid1  | <i>YPR061C</i> | Regulates Hsp70 ATPase activity; member of DnaJ family; involved in ER-associated degradation (ERAD)          |
|            | Scj1  | <i>YMR214W</i> | Regulates Hsp70 ATPase activity; ER luminal member of DnaJ family                                             |
|            | Hlj1  | <i>YMR161W</i> | Regulates Hsp70 ATPase activity; ER-membrane-associated member of DnaJ family                                 |
|            | Jem1  | <i>YJL073W</i> | Regulates Hsp70 ATPase activity; nuclear-membrane-associated member of DnaJ family                            |
|            | Sec63 | <i>YOR254C</i> | Regulates Hsp70 ATPase activity; member of DnaJ family; required for targeting and import of proteins into ER |
|            | Erj5  | <i>YFR041C</i> | Regulates Hsp70 ATPase activity; ER-associated member of DnaJ family                                          |
| Hsp70 NEFs | Sse1  | <i>YPL106C</i> | Cytosolic Hsp70 nucleotide exchange factor (Hsp110-type)                                                      |
|            | Sse2  | <i>YBR169C</i> | Cytosolic Hsp70 nucleotide exchange factor (Hsp110-type)                                                      |
|            | Fes1  | <i>YBR101C</i> | Cytosolic Hsp70 nucleotide exchange factor (HspBP1-type)                                                      |
|            | Snl1  | <i>YIL016W</i> | Cytosolic Hsp70 nucleotide exchange factor (Bag-1-type)                                                       |
|            | Lhs1  | <i>YKL073W</i> | ER luminal Hsp70 nucleotide exchange factor                                                                   |
|            | Sil1  | <i>YOL031C</i> | ER luminal Hsp70 nucleotide exchange factor (GrpE-type)                                                       |
| Small Hsps | Hsp42 | <i>YDR171W</i> | Forms barrel-shaped oligomers that suppress unfolded protein aggregation                                      |

|                      |       |                |                                                                                                                                             |
|----------------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                      | Hsp30 | <i>YCR021C</i> | Hydrophobic plasma membrane localized, negative regulator of the H(+)-ATPase Pma1p                                                          |
|                      | Hsp31 | <i>YDR533C</i> | Possible chaperone and cysteine protease with similarity to E. coli Hsp31                                                                   |
|                      | Hsp32 | <i>YPL280W</i> | Possible chaperone and cysteine protease with similarity to E. coli Hsp31                                                                   |
|                      | Hsp26 | <i>YBR072W</i> | Forms hollow, sphere-shaped oligomers that suppress unfolded proteins aggregation                                                           |
|                      | Hsp12 | <i>YFL014W</i> | Plasma membrane protein involved in maintaining membrane organization in stress conditions                                                  |
|                      | Hsp10 | <i>YOR020C</i> | Mitochondrial matrix co-chaperonin (GroES-like)                                                                                             |
| Cytosolic chaperonin | Cct1  | <i>YDR212W</i> | Alpha subunit of chaperonin-containing T-complex, which mediates protein folding in the cytosol; involved in actin cytoskeleton maintenance |
|                      | Cct2  | <i>YIL142W</i> | Subunit beta of the cytosolic chaperonin Cct ring complex                                                                                   |
|                      | Cct3  | <i>YJL014W</i> | Subunit of the cytosolic chaperonin Cct ring complex                                                                                        |
|                      | Cct4  | <i>YDL143W</i> | Subunit of the cytosolic chaperonin Cct ring complex                                                                                        |
|                      | Cct5  | <i>YJR064W</i> | Subunit of the cytosolic chaperonin Cct ring complex                                                                                        |
|                      | Cct6  | <i>YDR188W</i> | Subunit of the cytosolic chaperonin Cct ring complex                                                                                        |
|                      | Cct7  | <i>YJL111W</i> | Subunit of the cytosolic chaperonin Cct ring complex                                                                                        |
|                      | Cct8  | <i>YJL008C</i> | Subunit of the cytosolic chaperonin Cct ring complex                                                                                        |

**Table S3: Strains filtered-out of 37°C - 50°C results with significant death rates ( $p < 0.01$ ) due to low CFUs at time zero**

| ORF       | Name      | Death Rate | CFU (t = 0 minutes) |
|-----------|-----------|------------|---------------------|
| YPL193W   | RSA1      | -13.4      | 4.671               |
| YKL073W   | LHS1      | -12.86     | 9.342               |
| YKL081W   | TEF4      | -12.85     | 34.668              |
| YHR100C   | GEP4      | -9.17      | 4.2462              |
| YJR055W   | HIT1      | -8.7       | 5.0022              |
| YML099C   | ARG81     | -8.59      | 2.547               |
| YBR042C   | CST26     | -8.31      | 1.69848             |
| YOR290C   | SNF2      | -8.03      | 1.78668             |
| YBR119W   | MUD1      | -7.82      | 2.1222              |
| YOR299W   | BUD7      | -7.79      | 2.9718              |
| YNL208W   | YNL208W   | -7.53      | 2.1438              |
| YJL141C   | YAK1      | -7.3       | 1.42938             |
| YGR056W   | RSC1      | -7.12      | 10.1916             |
| YCR034W   | FEN1      | -7.01      | 12.7386             |
| YHR111W   | UBA4      | -6.86      | 19.53               |
| YMR021C   | MAC1      | -6.78      | 1.42938             |
| YKL212W   | SAC1      | -6.66      | 7.6428              |
| YIL128W   | MET18     | -6.48      | 10.6146             |
| YDR509W   | YDR509W   | -6.29      | 1.27386             |
| YIL110W   | MNI1      | -6.16      | 36.45               |
| YOR255W   | OSW1      | 6.19       | 1.07208             |
| YKR053C   | YSR3      | 6.24       | 0.3573              |
| YLR451W   | LEU3      | 6.31       | 30.564              |
| YJL204C   | RCY1      | 6.34       | 6.3684              |
| YMR166C   | YMR166C   | 6.51       | 0.3573              |
| YBL017C   | PEP1      | 6.6        | 0.42462             |
| YLR358C   | YLR358C   | 7.05       | 0.42462             |
| YLR068W   | FYV7      | 7.11       | 2.9718              |
| YJL144W   | YJL144W   | 7.13       | 0.3573              |
| YOL072W   | THP1      | 8.18       | 0.3573              |
| YPL078C   | ATP4      | 8.8        | 0.3573              |
| YDR523C   | SPS1      | 8.88       | 1.07208             |
| YMR154C   | RIM13     | 8.92       | 0.42462             |
| YLR325C   | RPL38     | 9.18       | 0.42462             |
| YOR376W   | YOR376W   | 10.52      | 0.3573              |
| YMR244C-A | YMR244C-A | 10.73      | 0.7146              |
| YCL025C   | AGP1      | 11.5       | 0.42462             |

**Table S4: 205 gene deletion strains present in MATa yeast deletion collection from OpenBiosystems (version 5), but absent in constructed prototrophic collection (due to auxotrophy, sterility, sporulation defects, or reasons unknown)**

| ORF       | Name      | ORF     | Name    | ORF     | Name    | ORF       | Name      |
|-----------|-----------|---------|---------|---------|---------|-----------|-----------|
| YAL047C   | SPC72     | YDR175C | RSM24   | YHR051W | COX6    | YLR231C   | BNA5      |
| YAR015W   | ADE1      | YDR194C | MSS116  | YHR057C | CPR2    | YLR292C   | SEC72     |
| YBL006C   | LDB7      | YDR195W | REF2    | YHR063C | PAN5    | YLR297W   | YLR297W   |
| YBL016W   | FUS3      | YDR200C | VPS64   | YHR071W | PCL5    | YLR299W   | ECM38     |
| YBL025W   | RRN10     | YDR216W | ADR1    | YHR081W | LRP1    | YLR300W   | EXG1      |
| YBL033C   | RIB1      | YDR227W | SIR4    | YHR147C | MRPL6   | YLR303W   | MET17     |
| YBL039C   | URA7      | YDR234W | LYS4    | YHR177W | YHR177W | YLR306W   | UBC12     |
| YBL058W   | SHP1      | YDR315C | IPK1    | YHR183W | GND1    | YLR307W   | CDA1      |
| YBL063W   | KIP1      | YDR410C | STE14   | YIL020C | HIS6    | YLR308W   | CDA2      |
| YBL072C   | RPS8A     | YDR433W | YDR433W | YIL052C | RPL34B  | YLR311C   | YLR311C   |
| YBL087C   | RPL23A    | YDR455C | YDR455C | YIL069C | RPS24B  | YLR322W   | VPS65     |
| YBL093C   | ROX3      | YDR461W | MFA1    | YIL084C | SDS3    | YLR341W   | SPO77     |
| YBR081C   | SPT7      | YDR495C | VPS3    | YIL094C | LYS12   | YLR362W   | STE11     |
| YBR085W   | AAC3      | YDR501W | PLM2    | YIL121W | QDR2    | YLR369W   | SSQ1      |
| YBR098W   | MMS4      | YEL004W | YEA4    | YIL125W | KGD1    | YLR420W   | URA4      |
| YBR101C   | FES1      | YEL029C | BUD16   | YIL154C | IMP2'   | YLR442C   | SIR3      |
| YBR112C   | CYC8      | YEL041W | YEF1    | YIL157C | COA1    | YLR447C   | VMA6      |
| YBR122C   | MRPL36    | YEL044W | IES6    | YIR009W | MSL1    | YMR091C   | NPL6      |
| YBR128C   | ATG14     | YEL072W | RMD6    | YIR021W | MRS1    | YMR194C-A | YMR194C-A |
| YBR213W   | MET8      | YER014W | HEM14   | YIR034C | LYS1    | YMR224C   | MRE11     |
| YBR293W   | VBA2      | YER026C | CHO1    | YJL071W | ARG2    | YMR299C   | DYN3      |
| YCL005W-A | VMA9      | YER055C | HIS1    | YJL080C | SCP160  | YNL059C   | ARP5      |
| YCL021W-A | YCL021W-A | YER069W | ARG5,6  | YJL088W | ARG3    | YNL138W   | SRV2      |
| YCL026C-B | HBN1      | YER070W | RNR1    | YJL102W | MEF2    | YNL197C   | WHI3      |
| YCL030C   | HIS4      | YER090W | TRP2    | YJL151C | SNA3    | YNL220W   | ADE12     |
| YCL032W   | STE50     | YER091C | MET6    | YJL176C | SWI3    | YNL277W   | MET2      |

|           |           |         |         |           |           |         |         |
|-----------|-----------|---------|---------|-----------|-----------|---------|---------|
| YCR002C   | CDC10     | YFL023W | BUD27   | YJR001W   | AVT1      | YNR006W | VPS27   |
| YCR020W-B | HTL1      | YFL026W | STE2    | YJR005C-A | YJR005C-A | YNR050C | LYS9    |
| YCR024C   | SLM5      | YFR008W | FAR7    | YJR010C-A | SPC1      | YNR068C | YNR068C |
| YCR053W   | THR4      | YFR025C | HIS2    | YJR010W   | MET3      | YOL058W | ARG1    |
| YCR087C-A | YCR087C-A | YFR030W | MET10   | YJR011C   | YJR011C   | YOL086C | ADH1    |
| YDL006W   | PTC1      | YGL012W | ERG4    | YJR018W   | YJR018W   | YOL143C | RIB4    |
| YDL021W   | GPM2      | YGL016W | KAP122  | YJR025C   | BNA1      | YOR128C | ADE2    |
| YDL038C   | #N/A      | YGL026C | TRP5    | YJR060W   | CBF1      | YOR198C | BFR1    |
| YDL041W   | YDL041W   | YGL031C | RPL24A  | YJR086W   | STE18     | YOR200W | YOR200W |
| YDL042C   | SIR2      | YGL033W | HOP2    | YJR104C   | SOD1      | YOR212W | STE4    |
| YDL053C   | PBP4      | YGL038C | OCH1    | YJR107W   | YJR107W   | YOR241W | MET7    |
| YDL073W   | YDL073W   | YGL101W | YGL101W | YJR109C   | CPA2      | YOR258W | HNT3    |
| YDL090C   | RAM1      | YGL125W | MET13   | YJR117W   | STE24     | YOR295W | UAF30   |
| YDL118W   | YDL118W   | YGL135W | RPL1B   | YJR122W   | IBA57     | YPL006W | NCR1    |
| YDL159W   | STE7      | YGL154C | LYS5    | YJR126C   | VPS70     | YPL049C | DIG1    |
| YDL160C-A | YDL160C-A | YGL206C | CHC1    | YJR139C   | HOM6      | YPL084W | BRO1    |
| YDL181W   | INH1      | YGL234W | ADE5,7  | YKL118W   | YKL118W   | YPL254W | HFI1    |
| YDL198C   | GGC1      | YGR167W | CLC1    | YKL135C   | APL2      | YPR032W | SRO7    |
| YDL225W   | SHS1      | YGR204W | ADE3    | YKL211C   | TRP3      | YPR036W | VMA13   |
| YDR007W   | TRP1      | YGR231C | PHB2    | YKL216W   | URA1      | YPR060C | ARO7    |
| YDR008C   | YDR008C   | YGR244C | LSC2    | YKR006C   | MRPL13    | YPR067W | ISA2    |
| YDR090C   | YDR090C   | YHL007C | STE20   | YKR069W   | MET1      | YPR099C | YPR099C |
| YDR103W   | STE5      | YHL025W | SNF6    | YLR027C   | AAT2      | YPR139C | VPS66   |
| YDR127W   | ARO1      | YHR005C | GPA1    | YLR059C   | REX2      |         |         |
| YDR138W   | HPR1      | YHR018C | ARG4    | YLR091W   | GEP5      |         |         |
| YDR158W   | HOM2      | YHR037W | PUT2    | YLR104W   | LCL2      |         |         |

**Table S5: Chemostat media components**

| <b>PHOSPHATE-LIMITED CHEMOSTAT MEDIA</b>        |                                  |
|-------------------------------------------------|----------------------------------|
| <b>Compound</b>                                 | <b>Amount per Liter of water</b> |
| KH <sub>2</sub> PO <sub>4</sub>                 | 10 mg (75 µM)                    |
| CaCl <sub>2</sub> x 2H <sub>2</sub> O           | 100 mg                           |
| MgSO <sub>4</sub> x 7H <sub>2</sub> O           | 500 mg                           |
| KCl                                             | 100 mg                           |
| (NH <sub>4</sub> ) <sub>2</sub> SO <sub>4</sub> | 5 g                              |
| NaCl                                            | 100 mg                           |
| Dextrose                                        | 5 g (0.5%)                       |
| 1000X vitamins                                  | 1 mL                             |
| 1000X metals                                    | 1 mL                             |
| <b>1000X VITAMINS</b>                           |                                  |
| Biotin                                          | 1 mg                             |
| Calcium pantothenate                            | 400 mg                           |
| Folic acid                                      | 2 mg                             |
| Myo-inositol                                    | 2 g                              |
| Nicotinic acid                                  | 400 mg                           |
| p-Aminobenzoic acid                             | 200 mg                           |
| Pyridoxine HCl                                  | 400 mg                           |
| Riboflavin                                      | 200 mg                           |
| Thiamine HCl                                    | 400 mg                           |
| <b>1000X METALS</b>                             |                                  |
| Boric acid                                      | 500 mg                           |
| Copper sulfate (x 5H <sub>2</sub> O)            | 40 mg                            |
| Potassium iodide                                | 100 mg                           |
| Ferric chloride (x 6H <sub>2</sub> O)           | 200 mg                           |
| Manganese sulfate (x H <sub>2</sub> O)          | 400 mg                           |
| Sodium molybdate (x 2H <sub>2</sub> O)          | 200 mg                           |
| Zinc sulfate (x 7H <sub>2</sub> O)              | 400 mg                           |

**Table S6: Colonies collected at each time-point**

| <b>Minutes</b> | <b>30°C - 50°C</b> | <b>37°C - 50°C</b> |
|----------------|--------------------|--------------------|
| 0              | $1.08 \times 10^6$ | $2.40 \times 10^6$ |
| 3              | $1.30 \times 10^6$ | $2.00 \times 10^6$ |
| 6              | $1.30 \times 10^6$ | $2.50 \times 10^6$ |
| 9              | $2.30 \times 10^6$ | $3.20 \times 10^6$ |
| 12             | $3.50 \times 10^6$ | $4.00 \times 10^6$ |
| 15             | $5.10 \times 10^6$ | $5.20 \times 10^6$ |

**Table S7: Barcode mapping and data filtering statistics**

|                                           | 30° C - 50° C                      |              |                       | 37° C - 50° C         |              |                       |
|-------------------------------------------|------------------------------------|--------------|-----------------------|-----------------------|--------------|-----------------------|
|                                           | Strains                            | % Collection | % Genome <sup>A</sup> | Strains               | % Collection | % Genome <sup>A</sup> |
| <b>Prototrophic Collection</b>            | 4783                               | 100          | 75.8                  | 4783                  | 100          | 75.8                  |
| <b>Mapped Barcodes<sup>B</sup></b>        | 3875                               | 81.1         | 61.4                  | 3910                  | 81.7         | 62.0                  |
| <b>≥ 40 CFU at t = 0</b>                  | 2954                               | 61.8         | 46.8                  | 3160                  | 66.1         | 50.1                  |
| <b>Significant Death Rate<sup>C</sup></b> | 208<br><i>175 neg.<br/>33 pos.</i> | 4.3          | 3.3                   | 65<br><i>all neg.</i> | 1.3          | 1.0                   |
| <b>Significant Death Rate<sup>D</sup></b> | 58<br><i>55 neg.<br/>3 pos.</i>    | 1.2          | 0.9                   | 10<br><i>all neg.</i> | 0.2          | 0.1                   |

A – 6,311 genes, according to GO-Slim Mapper (May 2011, [www.yeastgenome.org](http://www.yeastgenome.org))

B – Indicates how many strains had at least one barcode mapped for at least one time-point

C – Significant with  $p < 0.05$  (death rate  $\pm 3.54$  for 30-50;  $\pm 3.82$  for 37-50)

D – Significant with  $p < 0.01$  (death rate  $\pm 6.13$  for 30-50;  $\pm 6.16$  for 37-50)

**Table S8: GO Term Finder Analysis of Gene Sets****30°C – 50°C Negative Death Rate (p < 0.01; 55 genes)***No significant enrichment of any GO terms**23 out of 55 annotated to “biological process unknown”***30°C – 50°C Negative Death Rate (p < 0.05; 175 genes)***No significant enrichment of any GO terms**51 out of 175 annotated to “biological process unknown”***30°C – 50°C Positive Death Rate (p < 0.05; 33 genes)***No significant enrichment of any GO terms**9 out of 33 annotated to “biological process unknown”***37°C – 50°C Negative Death Rate (p < 0.01; 10 genes)***2 out of 10 annotated to “biological process unknown”*

| <b>Term (genes)</b>                                       | <b>Sample (%)<sup>a</sup></b> | <b>Pop. (%)<sup>b</sup></b> | <b>p-value</b>       |
|-----------------------------------------------------------|-------------------------------|-----------------------------|----------------------|
| Trehalose metabolic process<br>( <i>TPS2, HSP104</i> )    | 20.0                          | 0.3                         | $2.7 \times 10^{-2}$ |
| Disaccharide metabolic process<br>( <i>TPS2, HSP104</i> ) | 20.0                          | 0.5                         | $7.7 \times 10^{-2}$ |

**37°C – 50°C Negative Death Rate (p < 0.05; 65 genes)***12 out of 65 annotated to “biological process unknown”*

| <b>Term (genes)</b>                                                                               | <b>Sample (%)<sup>a</sup></b> | <b>Pop. (%)<sup>b</sup></b> | <b>p-value</b>       |
|---------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------|----------------------|
| Meiotic cell cycle<br>( <i>SIF2, RAD24, DMC1, HOS2, SET2, SET3, PMS1, MCK1, WTM2</i> )            | 14.1                          | 2.7                         | $1.3 \times 10^{-2}$ |
| M phase of meiotic cell cycle<br>( <i>SIF2, RAD24, DMC1, HOS2, SET2, SET3, PMS1, MCK1, WTM2</i> ) | 14.1                          | 2.7                         | $1.3 \times 10^{-2}$ |
| Meiosis<br>( <i>SIF2, RAD24, DMC1, HOS2, SET2, SET3, PMS1, MCK1, WTM2</i> )                       | 14.1                          | 2.7                         | $1.3 \times 10^{-2}$ |
| Negative regulation of meiosis<br>( <i>SIF2, HOS2, SET2, SET3</i> )                               | 6.2                           | 0.4                         | $3.6 \times 10^{-2}$ |
| Negative regulation of reproduction<br>( <i>SIF2, HOS2, SET2, SET3</i> )                          | 6.2                           | 0.5                         | $6.6 \times 10^{-2}$ |

a = percent of genes in the sample that fall into the indicated GO-Slim term

b = percent of genes in the 37°C - 50°C background gene set (3,160 genes)

The 30°C-50°C background gene set (2,954 genes) contained 34.7% genes of unknown function

The 37°C-50°C background gene set (3,160 genes) contained 34.1% genes of unknown function